

RIO FERDINAND

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A Bachelor of Applied Informatics Engineering from Universitas Airlangga, with a strong foundation in web development, machine learning, and system design. Experienced in building and optimizing web applications, implementing backend services, and developing AI-based solutions through both academic research and industry-focused projects. While my core strengths are in backend and machine learning development, I remain versatile and consistently broaden my expertise to perform effectively as a full-stack developer. I am eager to join a forward-thinking team where I can contribute my skills, embrace continuous learning, and help create impactful technology.

Experience

Badan Kesatuan Bangsa dan Politik Kota Surabaya - Surabaya, Indonesia

Oct 2025 – Feb 2026

Fullstack Developer - Contract

- Built and managed government web applications to move administrative tasks from paper-based to digital platforms.
- Developed a Digital Archive Management System featuring automated document retention policies using Laravel Schedulers and integrated PDF management.
- Built a Community Census and Database Platform to streamline the collection, verification, and processing of village-level data for government staff.
- Designed an Inventory Tracking System to manage internal department assets and resource allocation.
- Managed the full development lifecycle, ensuring all platforms met Kominfo security requirements before official deployment.
- Optimized relational database schemas to ensure data integrity and secure role-based access for multi-department use.

Lembaga Layanan Pendidikan Tinggi (LLDIKT) Wilayah VII - Surabaya, Indonesia

Sep 2024 – Dec 2024

Junior Backend Developer Intern

- Designed and implemented backend systems using Laravel, ensuring robust functionality and scalability.
- Developed and optimized database structures to guarantee data integrity, security, and system efficiency.
- Built and maintained APIs and business logic workflows to support lecturer data management, document submission, application status validation, and reporting.
- Integrated file upload and document management features to streamline administrative workflows.
- Collaborated with frontend developers to ensure seamless integration and user-centered performance.
- Debugged and maintained codebase, improving overall application stability and reliability.

PT SIEMENS INDONESIA - Jakarta, Indonesia

Feb 2024 – Jul 2024

Manufacturing Digitalization Intern

- Designed and implemented the frontend interface to ensure an intuitive and user-friendly experience.
- Developed robust backend functionality to handle procurement and maintenance workflows efficiently.
- Integrated database systems for secure and reliable data management.
- Conducted end-to-end testing to ensure seamless functionality across all features.
- Support digitalization project for checklist apps.

Education

UNIVERSITAS AIRLANGGA - Surabaya, Indonesia

Aug 2021 – Aug 2025

Bachelor of Applied Informatics Engineering, GPA: 3.77/4.00

- Awardee of Indonesian International Student Mobility Awards (IISMA) 2023 ([certificate](#))
- **Relevant Coursework:** Algorithm and Data Structure, Database Programming, Object-Oriented Programming, Information System Analysis & Design, Framework-based Web Development, UI/UX Design, Artificial Intelligence, Machine Learning

KANGWON NATIONAL UNIVERSITY ([certificate](#)) - Gangwon-do, Korea Selatan

Sep 2023 - Dec 2023

Exchange Student in AI & Software Engineering. GPA: 4.00/4.50

- **Course:** Digital Logic Design, Basic Optimization, Applied Math for AI, Business Communication Korean, Language for Foreigner

Laskar AI By Lintasarta - Jakarta, Indonesia (Remote)

Feb 2025 – Aug 2025

AI Engineer Cohort (Non-Degree Program)

- Engaging in an intensive independent study program focused on building a strong foundation in programming and data science.
- Learning essential programming concepts, Python basics, and introductory machine learning techniques.
- Exploring practical applications of data science through hands-on projects and real-world scenarios.

Projects

Food Image-Based Recipe Detection

Jun 2025 – Jul 2025

AI Engineer Cohort

- Designed and implemented an end-to-end data pipeline, including automated dataset partitioning (70:15:15) to prevent data leakage and ensure reliable evaluation.
- Fine-tuned MobileNetV2 using transfer learning and integrated a custom CNN head with Dropout layers to mitigate overfitting.
- Applied comprehensive data augmentation strategies and Early Stopping to improve generalization performance.
- Collaborated within a four-member team to develop and deploy a functional computer vision prototype for traditional Indonesian cuisine recognition.

Music Recommendation System Using Content-Based Filtering with Cosine Similarity

Dec 2024 – Jul 2025

Bachelor of Applied Informatics Engineering Thesis

- Built the system using a dataset of 16,000+ songs, analyzing intrinsic audio attributes such as tempo, valence, danceability, energy, and genre.
- Implemented a recommendation engine capable of delivering personalized suggestions based on similarity metrics.
- Evaluated system performance using Mean Opinion Score (MOS) and System Usability Scale (SUS).
- Achieved an average MOS score of 4.13/5, indicating high user satisfaction with recommendation quality.
- Attained a SUS score of 83.52/100, categorized as Excellent in usability.

Breast Cancer Classification

May 2025 – Jun 2025

AI Engineer Cohort

- Engineered a Machine Learning model for breast cancer detection using Random Forest and Recursive Feature Elimination (RFE) to identify and prioritize critical clinical features.
- Implemented a robust CI pipeline via GitHub Actions to automate code quality checks and model retraining workflows triggered by every commit.
- Integrated MLflow for experiment tracking, Docker for containerization, and utilized Prometheus and Grafana to monitor real-time system metrics, including request latency and prediction throughput.
- Deployed an interactive web application using Streamlit to provide end-users with intuitive visualizations of clinical data and diagnostic results.

Pharmacy App Development

Jun 2023 – Jul 2023

Backend Programmer

- Integrated WhatsApp API to enable seamless communication between customers and pharmacy staff.
- Optimized API performance to handle up to 10 concurrent users per hour.
- Designed and implemented database storage for chat history to ensure data persistence and accessibility.

Project Based Learning in Mobile Apps Coursework

Jan 2024 – Feb 2024

Android App Development

- Developed core app functionalities including login, signup, and product API integration.
- Designed the user interface for the app.
- Debugged and resolved issues to ensure smooth app performance.

Project Based Learning in UI Workshop Coursework

Jan 2024 – Feb 2024

Frontend Programmer

- Created a complete sales site interface including landing page, product pages, cart, and checkout.
- Ensured the website's responsiveness across various devices.
- Conducted cross-browser testing to ensure compatibility.

Additional Information

- **Skills:** Web Development, Database Management, Android Development, Machine Learning, MLOps, Deep Learning, Continuous Integration (CI), Data Analysis, Data Visualization
- **Programming Languages:** Python, Java, Dart, R, SQL, PHP, JavaScript, HTML, CSS
- **Frameworks & Tools:** TensorFlow, Keras, Laravel, Flutter, Streamlit, Docker, Grafana, Metabase, Figma, Git, GitHub
- **Language:** Indonesia (Native), English (Intermediate)
- **Certificate (2025):** TOEIC Test (Score: 935) - ([certificate](#))